# Building a Cross-Platform Back-End Application: The OmniCast Digital Broadcast Scheduling and Delivery System

**Abstract**

OmniCast is a multi-platform digital broadcast scheduling, real-time LiveStream and content delivery system developed at FPTU Quy Nhon AI Campus — FU Cahata for the PRN232 course under the supervision of Mentor Mr. Huỳnh Vang. It comprises a NestJS back end with 19 business modules and 140+ REST endpoints, a Next.js 15 web portal with 33 routes (including a 7-theme interactive LiveStream room, a 4-mode Live Studio console and a full administrative suite), a 31-screen Flutter mobile client, and a gRPC audit-logging microservice. Four principal architectural contributions are presented: (1) an end-to-end bi-directional LiveStream architecture combining WebRTC WHIP/WHEP, MediaMTX HLS and a Socket.IO `/live` gateway supporting live chat, floating reactions, 6-tier Super Chat donations, pinned broadcaster announcements and zero-downtime on-the-fly program switching; (2) a broadcast-schedule conflict-detection algorithm implemented as a single raw SQL query that computes the effective end time of an existing event; (3) a 24/7 EPG filler whose seeded selection guarantees that every channel-day grid is both complete and reproducible; and (4) an AI curation pipeline that issues a single large language model call, validates the output against a Zod schema, offers a heuristic fallback path and records full version provenance. The database schema comprises 21 Prisma models on PostgreSQL. The complete system has been deployed on Vercel and validated by 468 automated tests in archived CI run 40630318005: 163 backend, 228 web and 77 mobile tests, all of which passed.

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**FPTU QUY NHON AI CAMPUS (FU CAHATA)**
**FACULTY OF INFORMATION TECHNOLOGY — SOFTWARE ENGINEERING**
**COMPREHENSIVE PROJECT DELIVERABLE REPORT**
**COURSE: PRN232 — BUILDING CROSS-PLATFORM BACK-END APPLICATION**

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**OMNICAST: CROSS-PLATFORM DIGITAL BROADCAST SCHEDULING & STREAMING SYSTEM**
**Broadcast. Connect. Discover.**

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|---|---|
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| **Source Code Repository:** | https://github.com/TomOutfit/OmniCast_PRN232-PRM393_Project_Fa26Semester |
| **Live Environments:** | Web: https://omnicast-fe.vercel.app \| API: `/api/v1` \| MediaMTX WHIP/WHEP |
| **Codebase Scale & QA:** | **142,897 Lines of Code (LOC)** \| **468/468 Automated Tests Passed (100% CI)** |

broadcast scheduling, EPG, LiveStream, WebRTC WHIP/WHEP, Socket.IO, NestJS, Prisma, Next.js, gRPC, RBAC, Flutter, AI curation, conflict detection, FPTU Quy Nhon AI Campus, FU Cahata.

# Introduction

In recent years, the way Vietnamese audiences consume television and interactive live broadcasts (LiveStream) has changed dramatically. Viewers no longer sit in front of a screen at a fixed broadcast slot; they browse on-demand, join interactive live rooms with real-time chat and floating reactions, and expect content recommendations tailored to their interests. Broadcasters and digital-content providers, in turn, need a single, reliable and scalable scheduling and Live Studio control system that serves several delivery surfaces at once: the web, mobile applications and live-streaming devices.

This paper presents **OmniCast** — a multi-platform digital broadcast scheduling, real-time LiveStream and content delivery system developed at **FPTU Quy Nhon AI Campus — FU Cahata** for the course **PRN232 — Building Cross-Platform Back-End Application** under the academic mentorship of **Mr. Huỳnh Vang**. The system comprises a high-performance NestJS back end that serves both a Next.js 15 web portal (33 routes) and a Flutter mobile application (31 screens), with the core business logic concentrated on four capabilities: a bi-directional real-time LiveStream and Live Studio ecosystem, an electronic programme

guide (EPG) that is never empty, an atomic broadcast-schedule conflict-detection mechanism, and an AI editorial assistant with content-compliance validation.

## Motivation and Problem Statement

Four technical problems dominate the system's design, each derived from concrete project requirements:

1. **Multi-source LiveStream ingestion and real-time studio orchestration.** A modern broadcast platform must allow producers to go live from curated VOD libraries, device cameras/screens over WebRTC WHIP, or verified public catalogs, while supporting zero-downtime on-the-fly program switching without dropping viewers who are actively chatting, sending floating reactions and donating via Super Chat.
2. **Sparse schedules.** A real broadcast channel originates only a handful of productions per day. Building a 24-hour EPG grid directly from the event table leaves most of the grid empty and renders the day filter in the user interface meaningless. The system therefore requires an automatic filler that guarantees every channel-day grid is fully populated with valid content.
3. **Schedule conflicts.** If two programmes on the same channel overlap, viewers cannot tell which one is actually on air. The validation must detect all three overlap forms (head, tail, full containment) and must execute as a single database query to avoid race conditions.
4. **Manual content moderation.** The editorial workflow (writing descriptions, tagging, categorising) is manual and time-consuming. Automating it with a large language model reduces the load, but requires a fallback path when no API key is present, and the output must be schema-validated before it is persisted.

## Contributions

This paper makes the following technical contributions:

1. A layered **NestJS + Prisma + PostgreSQL** architecture with 19 business modules, 140+ REST endpoints and auto-generated OpenAPI documentation, protected by a four-role RBAC layer (`GUEST`, `VIEWER`, `STAFF`, `ADMIN`).
2. A **real-time LiveStream & Live Studio architecture** combining MediaMTX (WHIP/WHEP/HLS), Cloudinary Direct Upload and the Socket.IO `/live` gateway to support 4 studio broadcast modes, zero-downtime program switching, threaded live chat, 0ms floating emoji reactions, 6-tier Super Chat donations and pinned broadcaster announcements.

3. A broadcast-conflict detection algorithm implemented as a single raw SQL query that computes the effective end time with `COALESCE(endedAt, scheduledAt + duration)`.
4. A 24/7 EPG filler with seeded pseudo-random selection keyed on the (date, channel) pair, guaranteeing that the same day always renders the same schedule.
5. An AI editorial pipeline that issues a single consolidated model call (`gpt-4o`), validates the result against a Zod schema, and provides retry plus heuristic-fallback paths.
6. An audit-logging microservice communicating over gRPC (`audit_logger.proto`), deployed independently of the main API process.
7. A Next.js 15 web portal with 33 routes (including a 7-theme LiveStream lounge, a 4-tab Live Studio console and an AI Curator suite); archived CI run 40630318005 passed all 468 automated tests in its job scope.

## Project Team Organization and Role Allocation

The OmniCast project was executed at **FPTU Quy Nhon AI Campus — FU Cahata** under the supervision of Mentor **Mr. Huỳnh Vang**. Table 1 summarises the responsibilities of the four team members.

*The 4 project team members of OmniCast (FPTU Quy Nhon AI Campus — FU Cahata, Mentor: Mr. Huỳnh Vang).*

| Full Name | Role / ID | Primary Engineering Ownership |
|---|---|---|
| **Nguyễn Bình An** (`TomOutfit`) | Team Leader | NestJS 19-module core architecture, JWT & 4-role RBAC security, EPG conflict detection SQL, Direct Stream Engine, CI/CD & Release |
| **Trần Thanh Sơn** | `QE170080` | Standalone gRPC Audit Logger microservice (`audit_logger.proto`), Supabase PostgreSQL (21 Prisma models), Telemetry & Seeding |
| **Nguyễn Vĩnh Phú** | `QE180136` | Mobile Flutter Lead (31 screens, 12 BLoCs), `OmniPlayer` engine, `MobileStudioRecorderScreen` (WebRTC WHIP), Offline SQLite |
| **Đồng Hoàng Nguyên** (`teng122`) | Full-Stack & AI | Next.js 15 Cinema Dark Web Portal (33 routes), Socket.IO `/live` gateway, 7-theme Live Room, Live Studio Console & AI Curator module |

## Scope

This paper concentrates on the overall system architecture, the NestJS back end, the gRPC microservice, the database schema and the Next.js 15 web portal (PRN232 deliverables).

The 31-screen Flutter mobile application is presented in full detail in the companion PRM393 report.

# Related Work

This paper positions OmniCast against three main lines of work: back-end application frameworks, data-access protocols, and broadcast scheduling systems.

## Back-end Application Frameworks

NestJS adopts a decorator-driven dependency-injection model and a hierarchical module model, allowing each business domain (channels, programmes, users) to be packaged as an independent module with its own lifecycle. Compared with plain Express, the framework provides out-of-the-box URI versioning (`VersioningType.URI`), a `ValidationPipe` in `forbidNonWhitelisted` mode, an `ExceptionFilter` and an `Interceptor` — the four components the broadcasting problem demands so that every endpoint returns an identical response envelope and every input is validated strictly. Prisma serves as the data-access layer: the Prisma schema is the single source of truth and `prisma migrate` generates version-controlled SQL, so schema drift is tracked alongside the source code.

Two alternatives deserve comparison: FastAPI (Python) and ASP.NET Core (C#). FastAPI produces excellent OpenAPI documentation but the project has a fixed language budget; ASP.NET Core has a more mature ecosystem but also demands a language switch. NestJS allows the TypeScript toolchain to be reused with the Next.js portal , so a shared data type or an API contract change can propagate across both deliverables without a language boundary.

## Data-Access Protocols

Most legacy broadcasting systems expose their programme guide to mobile applications through OData , a REST extension with embedded query syntax ($filter, $orderby, $expand). OData is well suited when clients need flexible queries over a large catalogue without deploying server-side filtering logic. OmniCast takes the opposite approach: EPG queries have a fixed shape (by day, by channel, by time window), so plain REST endpoints with normalised query parameters are sufficient and simpler for both clients.

Compared with GraphQL , plain REST lowers the client-side cost because no additional runtime layer is needed on mobile, and it is easier to cache at the HTTP layer and to exercise from a command line. The trade-off is that every endpoint must be designed explicitly; in exchange, the entire API is documented with OpenAPI 3 using `@ApiOperation`, `@ApiResponse` and `@ApiQuery` decorators, so no endpoint is left undocumented.

## Broadcast Scheduling and AI Assistance

Commercial scheduling systems (such as EBU Radio-MEP and FTA EPG feeds) guarantee schedule integrity by applying relative-time conflict detection and substituting overrun programmes during live transmission. This is precisely the model OmniCast reproduces at a smaller scale: the slot of a programme is an interval $[t_0, t_1]$ on the timeline of a channel, and two overlapping intervals are invalid.

On the editorial-assistance side, commercial EPG systems have recently begun using large language models to generate descriptions and content tags . OmniCast's distinguishing feature is *output quality control*: every AI-generated report is validated against a Zod schema  before being written to the database and is stamped with the prompt version and the model version, so administrators can compare quality across prompt revisions.

## Gap Addressed by OmniCast

In summary, the gap this project fills is the absence of a complete, automatically verifiable broadcast-scheduling system at classroom scale: schedule-integrity guarantees, a full user experience across web and mobile, quality-controlled AI assistance, and the whole thing validated by 468 tests in CI run 40630318005: 163 backend, 228 web and 77 mobile tests in the workflow scope.

# System Architecture

## Layered Overview

The architecture shows the flow from both clients through the API to integrated services. The Next.js 15 portal (33 routes) and Flutter application (31 screens) use NestJS, with 19 business modules and 140+ REST endpoints. The backend connects to the Socket.IO real-time gateway, the gRPC audit microservice, MediaMTX, PostgreSQL/Supabase, Cloudinary and external APIs. MediaMTX also delivers HLS/WebRTC streams directly to clients.

*Overall layered architecture of OmniCast. Both the Next.js 15 web portal and the Flutter mobile application consume the same REST API and `/live` WebSocket gateway; audit logging is separated into a dedicated gRPC microservice.*

The notable design decision is a *single API process serving both clients*. The boundary between the web and mobile clients exists only at the transport layer, never at the business layer. A business rule is therefore written exactly once: for instance, the rule that "two programmes on the same channel must not overlap" is enforced in `ProgramsService.checkScheduleConflict` and behaves identically whether the schedule is created from the web administration portal or from the mobile application.

## End-to-End LiveStream & Live Studio Architecture

To satisfy ultra-low-latency live broadcasting and bi-directional interaction between studio producers (`STAFF/ADMIN`) and audiences (`VIEWER`), OmniCast implements a dedicated multi-pipeline live streaming architecture as shown in Fig. 2.

*End-to-end LiveStream and Live Studio architecture: combining 4 studio ingestion modes (Curated VOD, Camera/Screen WHIP WebRTC, Verified Public Catalog, Host Control Console) through `LiveStudioService`, MediaMTX and the Socket.IO `/live` gateway.*

The LiveStream architecture operates across three synchronised technical pillars:

1. **Four Studio Ingestion Pipelines:** Producers on Web (`/studio/live`) or Mobile (`LiveStudioScreen` & `MobileStudioRecorderScreen`) can initiate broadcasts across four modes: (i) *Curated Live* — selecting one or more recordings from the internal VOD library guarded by an Anti-Piracy filter; (ii) *Self-Record / Device Camera* — pushing real-time camera or screen-share streams via WebRTC WHIP to MediaMTX (port `:8889/whip`) or uploading local recordings directly to Cloudinary using signed parameters; (iii) *Verified Public Catalog* — one-click importing of verified public HLS/MP4 streams with server-side `HEAD` probing; and (iv) *Host Control Console* — broadcasting pinned announcements, moderating chat and switching active programmes on the fly.
2. **Multi-Protocol Transcoding & Delivery (MediaMTX & Direct Stream Engine):** Incoming WebRTC WHIP streams are remuxed by MediaMTX into WHEP and low-latency fMP4 HLS (`:8888/<streamKey>/index.m3u8`). Additionally, for Google Drive-hosted assets, the stream resolver rewrites share URLs to `drive.usercontent.google.com/download?id=...&confirm=t`, enabling direct hardware-accelerated streaming without Google account prompts.
3. **Real-Time Room Orchestration via Socket.IO (`/live`):** `LiveStreamGateway` manages per-event (`event:<eventId>`) and per-channel (`channel:<channelId>`) rooms, synchronising concurrent viewer counts (`live:viewerCount`), threaded chat (`live:chat:new`), 0ms floating emoji reactions (`live:reaction:new`), pinned host banners (`broadcaster:message`), Super Chat donations (`live:donation:received`) and zero-downtime stream switches (`live:program:changed`).

## NestJS Module Structure

`AppModule` registers 19 business modules alongside the infrastructure modules. Each module owns one business domain and exports the services it needs; controllers only call services and hold no business logic. Table 2 lists the actual back-end source layout.

*Back-end NestJS module structure.*

| Module | Primary responsibility | Controller |
|---|---|---|
| `auth` | Login, JWT, refresh, Google sign-in | `auth` |

| Module | Primary responsibility | Controller |
|---|---|---|
| channels | Broadcast channels, catalogue | channels |
| programs | Programmes, EPG, Live Studio, Social | programs |
| search | Cross-entity search | search |
| ai-curator | AI curation, statistics | ai-curator |
| audit-logger | Audit logging (gRPC) | audit-logger |
| watchlist | Watchlist | watchlist |
| users | User management | users |
| admin | Admin dashboard | admin |
| billing | Plans, payments | billing |
| notifications | Notifications, FCM push | notifications |
| newsletter | Newsletter subscription | newsletter |
| websocket | Realtime gateway /live | — |
| cloudinary | Media storage | cloudinary |
| prisma | Database access | — |
| health | Health probe | health |
| common | Filters, interceptors, Swagger | — |

## Protocol and Transport Integration

The back end serves four independent protocols, each serving a distinct group of needs:

- **REST/JSON over HTTP/1.1.** Paths carry the /api/v1 prefix and use VersioningType.URI. A middleware in main.ts automatically rewrites any /api/* request to /api/v1/*, so older clients never receive a 404 when the API version is upgraded.
- **WebSocket (Socket.IO /live).** The realtime gateway uses Socket.IO on namespace /live and exposes full live room orchestration events: event and channel subscribe/unsubscribe (subscribe:event, unsubscribe:event, subscribe:channel, unsubscribe:channel), concurrent viewer tracking (live:viewerCount), live chat (live:chat → live:chat:new), floating reactions (live:reaction → live:reaction:new), broadcaster pinned messages (broadcaster:message), message moderation (chat:delete → chat:deleted), on-the-fly program switching (live:program:changed), Super Chat donations (live:donation:received) and the WebRTC handshake (webrtc:offer, webrtc:answer, webrtc:ice-candidate).
- **gRPC.** The audit-logging microservice receives LogAudit, GetLogsByUser, GetLogsByAction, GetLogsByEntity and GetLogStats requests over audit_logger.proto.

- **WHIP/HLS/WebRTC.** MediaMTX ingests over WHIP and redistributes over HLS (LL-HLS, fMP4) and WHEP, configured in `backend/mediamtx/mediamtx.yml`.

### Database Connection and Audit Logging

The database is Supabase-managed PostgreSQL. One operational detail is worth noting: Supabase exposes two ports, `5432` for direct connections and `6543` for the transaction pooler. The `optimizeDatabaseUrl` helper in `PrismaService` automatically switches to `6543` and sets a suitable `connection_limit` when it detects the pooler, avoiding connection exhaustion when a serverless environment starts several instances concurrently.

Why a separate audit-logging microservice?

Audit logging has two properties the rest of the system does not share: a high write volume and a tolerance for delay. Writing straight into PostgreSQL would make every administrative operation wait on an extra write. Moving it into a gRPC microservice turns auditing into a non-blocking fire-and-forget operation: `AuditLoggerService` issues the RPC inside a non-awaited promise, while the microservice writes to the database and maintains index tables supporting queries by user, action, entity and aggregate statistics.

### External Integrations

Four external integrations are supported, all of which degrade gracefully when the API key is absent: YouTube Data API v3 (channel schedules), Twitch Helix (channel schedules), TMDB (film metadata enrichment) and OpenAI (content curation). This mechanism allows the whole system to run at full capability in a development environment without any third-party credential.

## Back-End Implementation

### JWT Authentication and Role-Based Access Control

The system authenticates with an access/refresh token pair. At login the password is verified with `bcrypt.compare` at 12 rounds (`BCRYPT_ROUNDS=12`); at registration it is hashed with `bcrypt.hash`. `AuthService` issues a 60-minute access token and a 90-day refresh token stored in the `RefreshToken` table. Token rotation is enforced: on every refresh the old token is revoked and a new one is issued, so a leaked token cannot be reused.

*Authentication and authorisation flow. The access token accompanies every request; `JwtAuthGuard` validates the signature and `RolesGuard` compares the role against the `@Roles` metadata.*

Two guards are implemented in `backend/src/auth/guards`. The `JwtAuthGuard` uses the Passport-JWT strategy and skips handlers marked `@Public()`. The `RolesGuard` reads `@Roles` metadata through the `Reflector`; a handler that declares no roles admits every authenticated user. A notable detail is that the guard normalises roles to string names

before comparing, so both the name form (`'STAFF'`) and the legacy numeric form (`'2'`) are accepted. The four roles are declared in the `UserRole` enum and ranked in the portal's `role-config.ts`.

*The four user roles and their permission scope.*

| Code | Value | Description |
|---|---|---|
| GUEST | 0 | Unauthenticated guest – public schedule and trailers only |
| VIEWER | 1 | Signed-in viewer – watchlist, comments, likes, watch history |
| STAFF | 2 | Content editor – manage channels, schedule, run AI Curator |
| ADMIN | 3 | Administrator – all STAFF rights plus user and audit-log management |

To enforce least privilege, sensitive operations are guarded at the controller level rather than in the user interface. For example, all eleven write endpoints in the `ai-curator` group and all broadcast control endpoints in `LiveStudioController` require `@Roles('STAFF','ADMIN')`, and every user-management or audit-log endpoint requires `@Roles('ADMIN')`.

## Broadcast Schedule Conflict Detection

The conflict-detection algorithm is the business core of the system. Each programme occupies a time interval $[\, t_0,\ t_0 + d\,]$ on the timeline of one channel. Two programmes collide if and only if the new interval starts before the existing one ends and ends after the existing one starts.

*The three detected overlap forms: head overlap, tail overlap and full containment. A schedule on a different channel causes no conflict.*

The hardest part is computing the *effective* end time of an existing event. The `endedAt` column can still be `NULL` for a `SCHEDULED` event, meaning the event has not actually finished. A naive Prisma `where` clause would skip such events, producing false negatives that allow conflicting schedules. The `checkScheduleConflict` helper therefore uses raw SQL to compute the effective end time with a `COALESCE/MAX` expression over `endedAt` and `scheduledAt + duration`.

```
SELECT id, title, "scheduledAt" AS scheduled_at, "duration"
FROM "LiveEvent"
WHERE "channelId" = $1::uuid
  AND status IN ('SCHEDULED', 'LIVE')
  AND "scheduledAt" < $2                      -- moi bat dau truoc khi cu ket
 thuc
  AND COALESCE(
        "endedAt",
        "scheduledAt" + (
          CASE
            WHEN "duration" IS NULL THEN ($3 || ' minutes')::interval
            WHEN "duration" > 1440  THEN ("duration" || ' seconds')::interval
```

```
            ELSE ("duration" || ' minutes')::interval
          END)
      ) > $4                                    -- moi ket thuc sau khi cu ba
t dau
```

The query only considers events in the `SCHEDULED` or `LIVE` states; `ENDED` and `CANCELLED` events cannot cause a conflict. The `CASE` expression also absorbs a unit inconsistency: the ingest layer writes `duration` in *seconds* while the EPG logic works in *minutes*, so values above 1440 are interpreted as seconds and converted, and everything else is treated as minutes. On a detected collision the service throws a `ConflictException` carrying HTTP status 409 together with the list of conflicting events, so the user interface can name the programme that caused the rejection.

## Live Studio Controller and Real-Time LiveStream Operations

The `programs` module exposes two dedicated controllers for live broadcasting: `LiveStudioController` (`backend/src/programs/live-studio.controller.ts`) and `SocialController` (`backend/src/programs/social.controller.ts`), working in tandem with `LiveStreamGateway` (`backend/src/websocket/live-stream.gateway.ts`). The complete live studio and viewer room lifecycle spans six technical operations:

1. **Mode 1 — Curated VOD Broadcast (`POST /programs/studio/curate-live`):** Allows producers (`STAFF`/`ADMIN`) to select one or more `recordingIds` from the channel or global VOD library for immediate broadcast. The service enforces an Anti-Piracy Guard (rejecting unauthorised YouTube/Twitch/Vimeo URLs), atomically ends any previous `LIVE` event on the same channel, creates the new `LiveEvent` with a multi-item `lineup`, and emits `live:started` over Socket.IO.

2. **Mode 2 — Self-Record, Signed Direct Upload & Webcam WHIP Live (`POST /programs/studio/self-record` & `/live/webcam/start`):** Supports two workflows: (a) *Direct-to-Cloudinary Signed Upload*: the client obtains signed parameters via `POST /programs/studio/recordings/sign`, uploads MP4/WebM directly to Cloudinary with live speed/ETA telemetry, and invokes `self-record` with server-side media probing to go live immediately (`IMMEDIATE`), schedule (`SCHEDULED`), or save as `PRIVATE` draft, complete with orphan asset cleanup (`DELETE /programs/studio/recordings/orphan`). (b) *WebRTC WHIP Direct Camera/Screen Streaming*: the client calls `POST /programs/studio/live/webcam/start` to allocate a `streamKey`, publishes camera or screen media via WebRTC WHIP to MediaMTX (`:8889/whip`), and serves low-latency HLS (`.m3u8`) to all viewers.

3. **Mode 3 — Verified Public Stream Catalog (`GET /programs/studio/public-catalog`):** Provides a curated catalog of royalty-free HLS and MP4 streams from Mux, Cloudinary and Internet Archive. Producers can filter by format, import with one click (`POST /programs/studio/recordings/library/from-catalog`), or register custom stream URLs with automated `HEAD` probing (`POST /programs/studio/recordings/library`).

4. **On-the-Fly Program Switching (`POST /programs/studio/events/:eventId/switch-program`):** When a live event contains multiple lineup items or a producer switches to another recording mid-broadcast, this endpoint updates the active `streamUrl` and broadcasts `live:program:changed` to room `event:<eventId>`. All connected viewers seamlessly transition to the new video stream without losing their chat session or WebSocket connection.

5. **Bi-Directional Room Interaction via `LiveStreamGateway` (`/live`):** Joining a room (`subscribe:event`) increments the in-memory and database-persisted viewer counter (`LiveEvent.viewerCount`) and broadcasts `live:viewerCount`. Viewers engage through live chat (`live:chat`), threaded replies (`parentId`), comment likes, and 0ms haptic floating emoji reactions (`HEART`, `FIRE`, `CLAP`, `WOW`/`PARTY`). Producers can broadcast pinned toast banners (`broadcaster:message`), delete inappropriate messages (`chat:delete` → `chat:deleted`), or stop the broadcast (`POST /programs/studio/live-events/:eventId/stop` with `keepAsVod`).

6. **PayOS Payment Gateway Integration & Super Chat Live Donations (`StreamDonation`):** Viewers in the live room can support creators across 6 preset gift tiers (Bronze 10,000 VND, Silver 20,000 VND, Gold 50,000 VND, Diamond 100,000 VND, Crown 200,000 VND, Elite 500,000 VND) with custom commendation messages. The system integrates directly with the PayOS payment gateway generating authentic MB Bank VietQR codes (Beneficiary: `NGUYEN BINH AN`, Account Number: `0935649095`), supporting one-touch mobile banking transfers or instant QR scanning. Upon transaction reconciliation via webhooks and status polling, the channel creator's wallet balance is automatically credited and a celebratory overlay is broadcast across the room via `live:donation:received`.

7. **Ultra-Low-Latency Delivery & Live Edge Synchronization:** The live broadcast pipeline combines WebRTC WHIP ingest with a tuned MediaMTX HLS packager delivering fine-grained 1-second segments. On client players (Web and Mobile), a two-stage adaptive speed catch-up mechanism maintains live synchronisation: when playback drift is between 1.8s and 4.0s, the player automatically boosts playback speed to 1.25x; if network congestion increases drift beyond 4.0s, an instant hard-seek forward to the live edge is executed; and as soon as drift returns under 900ms, playback returns seamlessly to 1.0x. Coupled with progressive fMP4 chunk decoding and post-buffering instant catch-up, viewers enjoy near real-time live playback with minimal latency.

## 24/7 EPG Filler Mechanism

Because channels originate only a handful of productions per day, the 24-hour EPG grid is materialised by `EpgScheduleFillerService`. The service guarantees five properties demanded by the design:

1. **No empty days.** Every channel-day grid covers the full 24 hours, so the day filter in the user interface always has content to render.

2. **Genre-aware selection.** Recordings whose `category` matches the channel category are preferred; the full library is used only when nothing matches.
3. **Episode splitting.** Recordings longer than `LONG_PROGRAM_MINUTES` are split into chunks of `EPISODE_MINUTES` so the grid exposes "Episode 1 / Episode 2 / …" entries instead of one massive block.
4. **Limited replay.** The same recording or episode may appear at most `MAX_REPEATS_PER_ITEM_PER_DAY` times in a single day, so viewers never see an episode loop endlessly.
5. **Reproducible randomness.** Selection uses a seeded PRNG keyed on the (date, channel) pair, so the same day always renders the same schedule while adjacent days surface different content.

The technically important point is that the service performs *no I/O*: it is pure logic that receives the recording list and the real programme list and returns the filled grid. As a result the 18 unit tests in `epg-filler.spec.ts` run without a database, and the whole scheduling layer becomes a pure function verifiable by properties.

## Content Ingestion and Enrichment

The ingestion layer collects schedules from YouTube, Twitch and RSS on a cron schedule (`0 */15 * * * *` for YouTube, `0 */30 * * * *` for Twitch), normalises them into a unified `NormalizedLiveEvent` structure and writes them to the `LiveEvent` table. An optional enrichment task then calls TMDB to supplement film metadata (poster, cast, rating) for events whose title matches.

*Ingestion flow from external sources. Every source is normalised into a single structure before being written to the database.*

## Quality-Controlled AI Curation Assistant

The `ai-curator` module is one of the most significant contributions. The original requirement was to automate writing descriptions, assigning categories and tagging programmes. The pipeline is designed around four principles:

1. **A single consolidated model call.** Prompt revision `curator-prompt-v4.0.0` merges three separate calls (description, tags, category) into one call with a JSON schema and few-shot examples, reducing both cost and latency to roughly a third.
2. **Zod schema validation.** The model output is validated against `ai-curator.schemas.ts`; when it does not validate, the request is routed to the heuristic fallback.
3. **Heuristic fallback.** When `OPENAI_API_KEY` is absent or equal to `dummy-key`, a local rule engine assesses sentiment, compliance and editorial tone to produce a structurally equivalent result.

4. **Version provenance.** Every report stores `promptVersion`, `aiModelVersion`, `openaiCallMs` and `openaiRetries`, allowing quality to be compared across prompt revisions.

*AI Curator pipeline: branching on API-key presence, Zod schema validation, and the heuristic fallback taken when validation fails.*

The hyper-parameters are pinned in `ai-curator.prompts.ts` (`temperature = 0.3`, `top_p = 0.95`), with an explicit instruction not to change `temperature` without re-evaluating the few-shot examples. The versioning convention is semantic: MAJOR for schema changes, MINOR for rewording that preserves the JSON shape, PATCH for copy edits. The statistics module `ai-curator-stats.service.ts` aggregates by `broadcastSuitability` group, by model version and by time period.

## Search, Billing and Supporting Modules

The `search` module performs cross-entity search: channels, programmes and recordings are queried simultaneously with case-insensitive `contains` conditions, combined with category filtering and pagination. Results are merged into a single object carrying the total result count, matching the single-page search layout that renders grouped result sections.

The `billing` module manages plans and payment history; the `notifications` module pushes notifications through Firebase Cloud Messaging based on the `DeviceToken` table; the `newsletter` module manages subscriptions; and the `cloudinary` module handles image and video uploads for the live-recording flow in the studio.

## Cross-Cutting Technical Conventions

Four conventions applied across the whole system deserve explicit mention because they determine API consistency:

1. **Uniform response envelope.** A global `TransformInterceptor` wraps every result in `{success, statusCode, data, meta, timestamp}`, so clients need exactly one decoding path.
2. **Centralised exception handling.** `HttpExceptionFilter` normalises every error into the same structure, including user-facing Vietnamese messages.
3. **Strict input validation.** A `ValidationPipe` with `whitelist` and `forbidNonWhitelisted` strips every field not declared in the DTO, preventing mass assignment.
4. **Generated API documentation.** Swagger decorators on each controller produce OpenAPI documentation at `/swagger`, covering 19 endpoints in the `ai-curator` group together with `channels`, `programs` and the remaining groups.

# The Next.js 15 Web Portal

## App Router Architecture

The portal is built on Next.js 15 with the App Router, React 19 and Tailwind CSS with a “Cinema Dark” theme. The App Router replaces the `pages/` model with nested *Server Components* and *Server Actions*, allowing data to be fetched directly on the server without an intermediate API layer.

LIVE FEED #01 SATELLITE DOWNLINK // 2160p60 HEVC | LATENCY: 0.42s | BITRATE: 34.8 Mbps

Xem lại từ đầu (DVR) | Multi-View (4 Kênh) | Chế độ Rạp chiếu

Trận Chung Kết Liên Minh Huyền Thoại Mùa Xuân #2

Omni Indie Games LIVE | Trận Chung Kết Liên Minh Huyền Thoại Mùa Xuân #2

RMA 2 : 2 MCI 78:42 2ND HALF • ET POSS

GÓC QUAY CHỦ ĐỘNG (4): Main (Omni Academy) | Spider-Cam (Omni Esports) | Góc Cam #3 (Omni Travel VN) | Góc Cam #4 (Omni Health)

Bắt đầu phát sóng: 02:00:00

-00:11:18 (TIME-SHIFT 4H) | SÓNG (LIVE HEAD) 03:48:42

Bình luận: TV 1 (Quang Huy & Anh Ngọc) | DOLBY ATMOS 5.1 | 4K UHD 60FPS

Live Chat 12.4K | Thống Kê | Bình Chọn MVP

OmniBot: Chào mừng 124,000 người xem cùng theo dõi trận chung kết!

Trần Hoàng VIP MEMBER 03:46
Pha cứu thua không tưởng của Courtois!! Man City ép sân nghẹt thở quá 🔥🔥

Lê Khánh 03:47
Foden trận này đá cánh hay nhất mùa rồi. Góc cam Tactical xem rõ vị trí chạy chỗ đã mắt.

Văn Minh MOD 03:47
Đề nghị các bạn giữ lịch sự trong box chat nhé. Chuẩn bị có hiệp phụ!

An Nguyên 03:48
Chất lượng 4K mượt đét không giật lag tí nào, âm thanh sân Wembley nghe như đang ngồi khán đài luôn ấy 💗

Sút | Cháy | Tuyệt | Đau tim | Cup

Bình luận về trận đấu...

ĐỘC QUYỀN TRUYỀN HÌNH Kênh: Omni Indie Games • 19,963 Khán giả

## Trận Chung Kết Liên Minh Huyền Thoại Mùa Xuân #2

Chia sẻ | 148.2K | Phát lên TV

Trận Chung Kết Liên Minh Huyền Thoại Mùa Xuân #2 - phát sóng trực tiếp trên Omni Indie Games. Đừng bỏ lỡ chương trình hấp dẫn này!

EPG REALTIME SYNC UTC+7 (HANOI)

## HỆ THỐNG 25 KÊNH Lịch Phát Sóng Điện Tử & Catch-Up 7 Ngày

HIỆN TẠI (LIVE) | Tìm tên trận đấu, phim, gameshow... | Omni Originals | Xuất Lịch

CATCH-UP: T.SÁU 25/09 | T.BẢY 26/09 | CN 27/09 | T.HAI 28/09 | T.BA 29/09 | HÔM QUA 30/09 | HÔM NAY 01/10 | NGÀY MAI 02/10 | T.BẢY 03/10 | CN 04/10 | T.HAI 05/10 | T.BA 06/10 | T.TƯ 07/10 | T.NĂM 08/10

TẤT CẢ (25 KÊNH) | THỂ THAO & ESPORTS (3) | PHIM & DRAMA (2) | SHOWS & NHẠC (5) | TRI THỨC & CÔNG NGHỆ (8) | ĐỜI SỐNG & DU LỊCH (7)

Sáng (06:00-12:00) | Chiều (12:00-18:00) | • Giờ Vàng (18:00-23:00) | Đêm (23:00-06:00)

Tính năng Catch-Up 7 ngày hỗ trợ toàn bộ 25 kênh cho phép xem lại chương trình đã phát sóng với âm thanh Dolby 5.1 và phụ đề đa ngôn ngữ. | Đã phát (Xem lại) | Đang phát (Live) | Sắp phát sóng

| 25 KÊNH PHÁT SÓNG CH #001 - #025 | 18:00 | 18:30 | 19:00 | 19:30 | 20:00 | 20:30 | 21:00 | 21:30 | 22:00 |
|---|---|---|---|---|---|---|---|---|---|
| CH #001 Omni Academy EDUCATION | 07:00 - 08:30 REPLAY HD Workshop Kỹ Năng Thuyết Trì... | | 08:30 - 10:00 REPLAY HD Workshop Kỹ Năng Thuyết Trì... | | 10:00 - 11:30 REPLAY HD Ôn Thi Học Kỳ 2 - Môn Ngữ Vă... | | 11:30 - 13:00 REPLAY HD Ôn Thi Học Kỳ 2 - Môn Ngữ Vă... | | 13:00 - 14:30 REPLAY HD Hướng Dẫn Làm Đồ Ăn Tốt... |
| CH #002 Omni Art & Design ART | 07:00 - 08:30 REPLAY HD Portrait of a Woman — Quinten... | | 08:30 - 10:00 REPLAY HD Portrait of a Woman — Quinten... | | 10:00 - 11:30 REPLAY HD Chrysanthemums in the Garde... | | 11:30 - 13:00 REPLAY HD Chrysanthemums in the Garde... | | 13:00 - 14:30 REPLAY HD Phỏng Vấn Illustrator Việt Nam... |
| CH #003 Omni Audiobook PODCAST | 07:00 - 08:30 REPLAY HD Tọa Đàm Giáo Dục: Con Đường... | | 08:30 - 10:00 REPLAY HD Tọa Đàm Giáo Dục: Con Đường... | | 10:00 - 11:30 REPLAY HD Sách Nói Tiếng Anh: Atomic... | | 11:30 - 13:00 REPLAY HD Sách Nói Tiếng Anh: Atomic... | | 13:00 - 14:30 REPLAY HD Talk Show: Chuyện Chưa Kể C... |
| CH #004 Omni Business BUSINESS | 07:00 - 08:30 REPLAY HD Phỏng Vấn Shark Tank: Đánh ... | | 08:30 - 10:00 REPLAY HD Phỏng Vấn Shark Tank: Đánh ... | | 10:00 - 11:30 REPLAY HD Bài Học Khởi Nghiệp Từ Các Tỷ... | | 11:30 - 13:00 REPLAY HD Bài Học Khởi Nghiệp Từ Các Tỷ... | | 13:00 - 14:30 REPLAY HD Hướng Dẫn Xây Dựng Thương... |
| CH #005 Omni Cine CINE | 07:00 - 08:30 REPLAY HD Phim Ngắn: "Ánh Sáng Nơi Cuố... | | 08:30 - 10:00 REPLAY HD Phim Ngắn: "Ánh Sáng Nơi Cuố... | | 10:00 - 11:30 REPLAY HD Omni Cine — Đang phát sóng | | 11:30 - 13:00 REPLAY HD Omni Cine — Đang phát sóng | | 13:00 - 14:30 REPLAY HD Omni Cine — Đang phát sóng |

Mỗi ô: 5.1 SURROUND | Multi-Cam Ready | Xem Ngay

### EPG Realtime & Catch-Up 7 Ngày

Xem lại trọn vẹn toàn bộ các sự kiện thể thao, phim truyện và show truyền hình đã phát sóng với phụ đề đa ngôn ngữ.

### Multi-Cam 4K UHD 60FPS

Tự do chuyển đổi góc quay camera Tactical, Spider-cam hoặc theo dõi riêng biệt từng ngôi sao cầu thủ yêu thích.

### Dolby Atmos & Đa Luồng Bình Luận

Âm thanh vòm vòm sống động như tại khán đài, lựa chọn bình luận viên quốc tế hoặc tiếng Việt theo ý muốn.

### AI Curator Studio Tự Động

Trợ lý AI tự động sắp xếp lịch phát sóng, tối ưu hóa khung giờ vàng và gợi ý nội dung cá nhân hóa.

NETWORK STATUS: ALL 25 CHANNELS BROADCASTING NOMINAL (4K HDR 60FPS) | SATELLITE SYNC: UTC+7 (HANOI) | DOLBY ATMOS 5.1 CERTIFIED | HEVC / H.265 LOW LATENCY

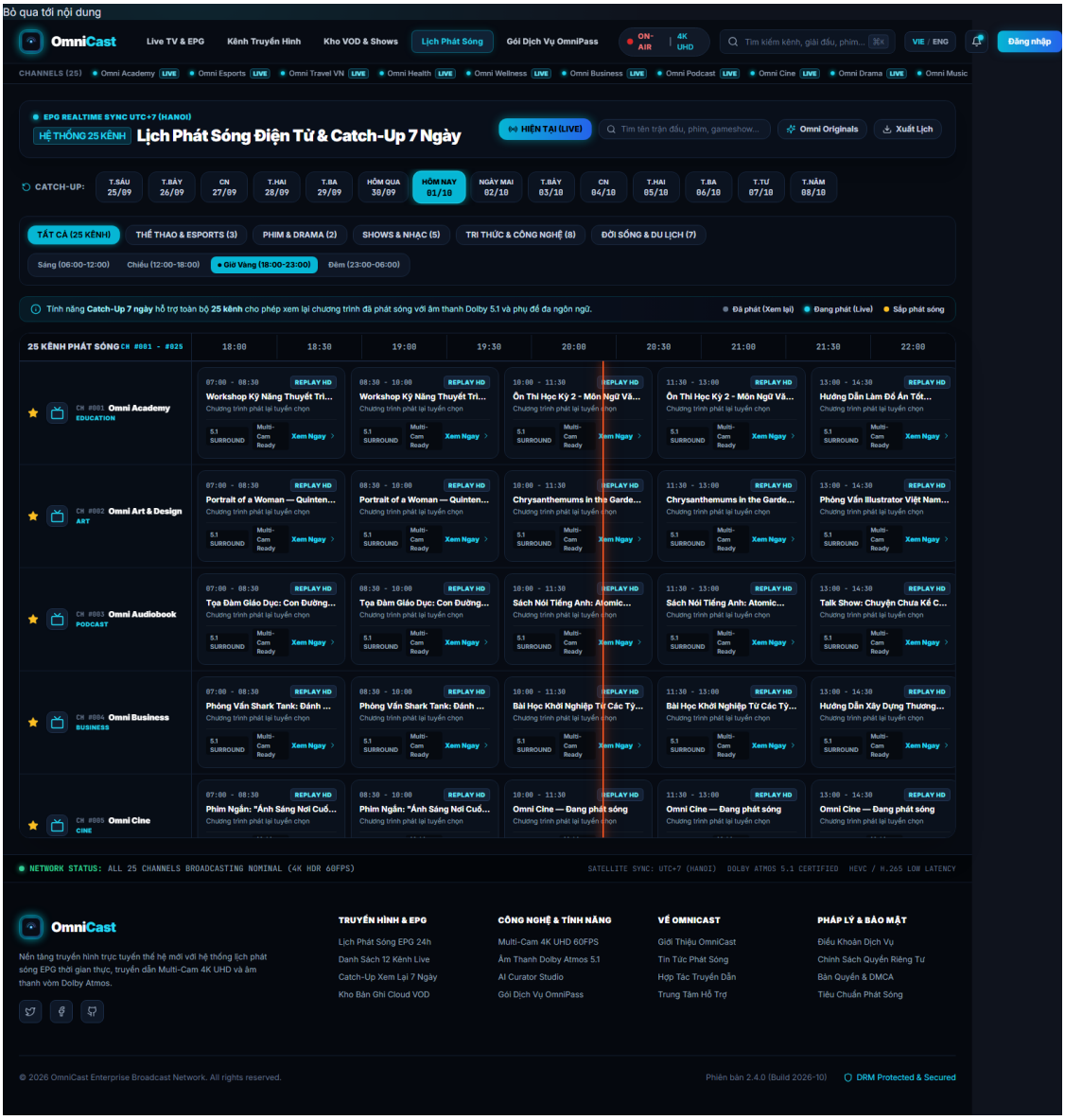
*The web portal home page on desktop: featured carousel, channel grid by category and the role-aware navigation bar.*

## Complete 33-Route Web Portal Inventory

The web portal exposes 33 routes generated from 24 directories under `frontend/app/`. Table [tab:web_routes] catalogues all 33 screens across five functional clusters: (1) Public Discovery & EPG; (2) Interactive LiveStream & VOD Rooms; (3) Live Studio Console & AI Curator Suite (`STAFF/ADMIN`); (4) Account, Personalisation & Administration (`ADMIN`); and (5) Legal, Compliance & Broadcast Standards.

| Cluster | Route Path (33) | RBAC Role | Key Screen Features & Interactive Operations |
|---|---|---|---|
| **1. Public Discovery & EPG** | `/`, `/channels`, `/channels/[id]` | Public | Hero spotlight carousel, category-filtered channel grid, channel detail page with live stream & EPG. |
| | `/epg` | Public | Zero-gap 24/7 EPG timeline grid, real-time Now-Line indicator, date and channel filters. |
| | `/programs`, `/programs/[id]` | Public | Programme directory, programme detail page with production tags, upcoming airings and Watchlist button. |
| | `/recordings`, `/recordings/[id]` | Public | Published VOD recording library, multi-engine replay player with comments and related VODs. |
| | `/search` | Public | Cross-entity search (Channels, Programmes, VOD Recordings) with keyword and genre filters. |
| | `/live` | Public | Live Hub: auto-refreshes every 15s, displays active LIVE event & total CCU pills, sorts by trending viewers or newest start time. |
| | `/live/[id]` | Public / Viewer | Interactive LiveStream Room: 7 genre-themed lounges, `JoinLobbyCard` prompt, Cinema Mode, lineup program switcher, real-time Socket.IO |

| Cluster | Route Path (33) | RBAC Role | Key Screen Features & Interactive Operations |
|---|---|---|---|
| | | | chat, 4 floating emoji reactions, pinned host banner. |
| **3. Live Studio & AI Curator** | `/studio`, `/studio/live` | `STAFF+` | 4-Tab Live Broadcast Console: (1) Curated VOD Live; (2) Camera/Screen WHIP WebRTC & Signed Cloudinary Dropzone; (3) Verified Public Catalog; (4) Host Announcement & Stop Broadcast. |
| | `/studio/curator` | `STAFF+` | AI Editorial Workbench (`gpt-4o` + heuristic fallback): sentiment scoring, broadcast compliance check, auto-tagging. |
| | `/studio/curator/audit`, `/studio/curator/dashboard` | `STAFF+` | Prompt-version provenance audit log (`promptVersion`), suitability breakdown and latency analytics. |
| | `/login`, `/register`, `/forgot-password`, `/reset-password` | Public | JWT + Google OAuth2 sign-in, account registration, tokenised password recovery. |
| | `/profile`, `/settings`, `/watchlist`, `/notifications` | `VIEWER+` | User profile, VIP plan management, cross-device watchlist, notification centre and global session revocation (`logout-all`). |
| | `/admin`, `/admin/users`, `/admin/audit-logs` | `ADMIN` | System telemetry overview, user RBAC management, gRPC audit log explorer by action/entity. |
| **5. Legal & Support** | `/terms`, `/privacy`, `/dmca`, `/standards`, `/broadcasting-standards`, `/help`, `/docs` | Public | Terms of service, privacy policy, DMCA copyright policy, community & broadcasting standards, help centre and docs. |




*The electronic programme guide page (/epg) with the time-slot grid and the day selector.*

## Interactive LiveStream Room (`/live` & `/live/[id]`) and Viewer Operations

The web LiveStream experience is engineered as a full-featured interactive broadcast lounge across two core screens:

1. **Live Broadcast Hub & Live Stream Schedule (`/live` & Home):** Polls active `LIVE` events every 15 seconds, displays real-time header badges for total active broadcasts and aggregate concurrent viewers (CCU), integrates an interactive **LiveStream Broadcast Schedule** with 4 instant filter tabs (*All*, *Live Now*, *Golden Hour 19h00–22h30*, *Today*), supports sorting by viewer count or start time, and surfaces a direct “Go to Live Studio” shortcut for `STAFF/ADMIN` users.

2. **Interactive Lobby & 7 Genre-Themed Broadcast Lounges (`/live/[id]`):** Upon entering a stream, viewers encounter a `JoinLobbyCard` previewing the stream status and live viewer count before entering the room. Once inside, `resolveLiveRoomTheme` dynamically styles the entire room into one of **7 genre-specific themes**: *Kids Wonderland*, *Podcast & Audiobook*, *Music Live Stage*, *Academy & Tech*, *Sports Arena*, *Obsidian Theater* (Cinema/Drama) and *Broadcast Lounge* (Default).
3. **Low-Latency Multi-Engine Player, Cinema Mode & Instant Lineup Switching:** `VideoPlayer` seamlessly handles MediaMTX HLS/WebRTC streams (`cookieCheck=1`), direct Google Drive streams and YouTube embeds. The player is optimized with progressive fMP4 chunk decoding (`progressive: true`), fine-tuned buffer sync (`liveSyncDurationCount: 1`) and adaptive 1.25x speed catch-up when drift exceeds 1.8s. Viewers can toggle *Cinema Mode* (lights-off focus) to dim surrounding chrome and expand the stage to full width. When a broadcast contains multiple lineup items or is controlled by `STAFF/ADMIN`, the lineup bar triggers `switchProgramOnLiveEvent`, switching the active video stream for all viewers in the room without page reload.
4. **Real-Time Socket.IO (`/live`) Operations & PayOS Donations:** The live interaction sidebar connects to the `/live` namespace, enabling viewers to: (i) post real-time chat messages with auto-scroll; (ii) fire 4 floating emoji reactions (`HEART`, `FIRE`, `CLAP`, `WOW`) that animate upward over the video canvas with optimistic counter updates; (iii) send Super Chat live donations via `LiveDonationDialog` integrated with the **PayOS** gateway and genuine **MB Bank** VietQR generation (Beneficiary: `NGUYEN BINH AN`, Account Number: `0935649095`) supporting 6 gift tiers; and (iv) receive pinned host announcements (`broadcaster:message`) rendered as a prominent banner across the stage.

## Web Live Studio Broadcast Console (`/studio/live`)

The `/studio/live` route (`frontend/app/studio/live/page.tsx`) provides a production-grade broadcast control console for `STAFF` and `ADMIN` operators across four tabs:

- **Tab 1 — Curated Live:** Select the target channel, configure broadcast metadata, and pick one or more VOD recordings from the channel or global library to launch a multi-item `lineup`.
- **Tab 2 — Self-Record / File Upload / Webcam WHIP Live (`WebcamLiveBroadcaster` & `WebStudioViewfinder`):** Supports browser camera capture, screen sharing (`getDisplayMedia`), direct WebRTC WHIP publishing to MediaMTX, or drag-and-drop file ingestion via `UploadDropzone` + `UploadQueue` (signed direct-to-Cloudinary upload with live MB/s speed, ETA, automatic HTML5 `canvas` thumbnail extraction, and 3 visibility modes: `IMMEDIATE`, `SCHEDULED`, `PRIVATE`).

- **Tab 3 — Verified Public Catalog:** Browse verified HLS/MP4 streams with format filters and one-click channel library import.
- **Tab 4 — Live Host Console:** Send pinned broadcaster announcements (`broadcaster:message`) to all connected viewers and terminate active broadcasts (`stopLiveEvent`) with the `keepAsVod` archival option.

## Role-Aware Interface and AI Curator Suite

A single configuration file, `frontend/lib/constants/role-config.ts`, determines which components appear for each role. The `hasMinimumRole` helper compares role ranks, so promoting a capability from `VIEWER` to `STAFF` is a one-line edit. The `/studio/curator` page is the AI curation interface: it lists programmes awaiting review, allows re-running the AI pipeline, and exposes the confidence score, broadcast-suitability classification and compliance flags. The `/studio/curator/audit` page presents the run history with prompt-version provenance, while `/studio/curator/dashboard` aggregates statistics by category and broadcast suitability.

BROADCAST NETWORK // 12 HIGH-BITRATE CHANNELS

# Danh Sách Kênh Truyền Hình Trực Tuyến

Hệ thống kênh độc quyền phát sóng 24/7 với độ trễ siêu thấp, chuẩn nén HEVC 2160p60 và hỗ trợ xem lại Catch-up 7 ngày.

Tìm kiếm kênh hoặc chương trình...

Tất Cả (25 Kênh) | Thể Thao | Điện Ảnh 4K | Phim Truyện | Show & Reality | Tin Tức 24/7 | Âm Nhạc | Thiếu Nhi | Công Nghệ & AI | Ẩm Thực | Khám Phá | Esports & Gaming | Podcast & Audio | Giáo Dục | Đời Sống & Fashion | Du Lịch | Nghệ Thuật | Kinh Doanh | Sức Khỏe

CH #001 · LIVE
**Omni Academy**
GIÁO DỤC
Đang phát: ON-AIR 4K
Workshop Toán Tư Duy Cho Trẻ Em 8-12 Tu...
▶ Xem Trực Tiếp

CH #002 · LIVE
**Omni Esports**
TRÒ CHƠI
Đang phát: ON-AIR 4K
Tournament Valorant Champions Tour 202...
▶ Xem Trực Tiếp

CH #003 · LIVE
**Omni Travel VN**
DU LỊCH
Đang phát: ON-AIR 4K
Khám Phá Hang Sơn Đoòng - Hang Động L...
▶ Xem Trực Tiếp

CH #004 · LIVE
**Omni Health**
SỨC KHỎE
Đang phát: ON-AIR 4K
Tư Vấn Dinh Dưỡng Cho Trẻ Biếng Ăn Từ...
▶ Xem Trực Tiếp

CH #005 · LIVE
**Omni Wellness**
PHONG CÁCH SỐNG
Đang phát: ON-AIR 4K
Yoga Buổi Sáng 30 Phút Cho Người Mới Bắt...
▶ Xem Trực Tiếp

CH #006 · LIVE
**Omni Business**
KINH DOANH
Đang phát: ON-AIR 4K
Phân Tích Thị Trường Chứng Khoán Tuần...
▶ Xem Trực Tiếp

CH #007 · LIVE
**Omni Podcast**
PODCAST
Đang phát: ON-AIR 4K
Phỏng Vấn Độc Quyền CEO Công Ty Công...
▶ Xem Trực Tiếp

CH #008 · LIVE
**Omni Cine**
ĐIỆN ẢNH
Đang phát: ON-AIR 4K
Omni Cine — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #009 · LIVE
**Omni Drama**
PHIM TRUYỆN
Đang phát: ON-AIR 4K
Omni Drama — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #010 · LIVE
**Omni Music**
ÂM NHẠC
Đang phát: ON-AIR 4K
Omni Music — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #011 · LIVE
**Omni News**
TIN TỨC
Đang phát: ON-AIR 4K
Tin Tức 24H: Cập Nhật Tình Hình Thế Giới
▶ Xem Trực Tiếp

CH #012 · LIVE
**Omni Show**
SHOW
Đang phát: ON-AIR 4K
Omni Star Talk: Gặp Gỡ Dàn Diễn Viên Bom...
▶ Xem Trực Tiếp

CH #013 · LIVE
**Omni Sport 1**
THỂ THAO
Đang phát: ON-AIR 4K
Trực Tiếp: Trận Siêu Kinh Điển Champions...
▶ Xem Trực Tiếp

CH #014 · LIVE
**Omni Sport 2**
THỂ THAO
Đang phát: ON-AIR 4K
Omni Sport 2 — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #015 · LIVE
**Omni Discovery**
KHÁM PHÁ
Đang phát: ON-AIR 4K
Omni Discovery — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #016 · LIVE
**Omni Entertain**
GIẢI TRÍ
Đang phát: ON-AIR 4K
Omni Entertain — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #017 · LIVE
**Omni Tech**
CÔNG NGHỆ
Đang phát: ON-AIR 4K
Omni Tech — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #018 · LIVE
**Omni Fashion**
PHONG CÁCH SỐNG
Đang phát: ON-AIR 4K
Hướng Dẫn Thiền Định Mindfulness Trong ...
▶ Xem Trực Tiếp

CH #019 · LIVE
**Omni Travel World**
DU LỊCH
Đang phát: ON-AIR 4K
Vlog Du Lịch Bali 5 Ngày 4 Đêm Dưới 15 Tri...
▶ Xem Trực Tiếp

CH #020 · LIVE
**Omni Skill Lab**
GIÁO DỤC
Đang phát: ON-AIR 4K
Workshop Toán Tư Duy Cho Trẻ Em 8-12 Tu...
▶ Xem Trực Tiếp

CH #021 · LIVE
**Omni Indie Games**
TRÒ CHƠI
Đang phát: ON-AIR 4K
Tournament Valorant Champions Tour 202...
▶ Xem Trực Tiếp

CH #022 · LIVE
**Omni Art & Design**
NGHỆ THUẬT
Đang phát: ON-AIR 4K
Behance Review: 10 Portfolio Thiết Kế Ấn...
▶ Xem Trực Tiếp

CH #023 · LIVE
**Omni Audiobook**
PODCAST
Đang phát: ON-AIR 4K
Phỏng Vấn Độc Quyền CEO Công Ty Công...
▶ Xem Trực Tiếp

CH #024 · LIVE
**Omni Food**
ẨM THỰC
Đang phát: ON-AIR 4K
Omni Food — Chương trình đặc biệt
▶ Xem Trực Tiếp

CH #025 · LIVE
**Omni Kids**
THIẾU NHI
Đang phát: ON-AIR 4K
Omni Kids — Chương trình đặc biệt
▶ Xem Trực Tiếp

*The channel catalogue page (/channels) with category filters and content badges.*

### Client-Side Security, Session Handling and Testing

The portal uses `next-themes` for light/dark support and a `safe-log` layer (`redact.ts`) that masks sensitive fields (passwords, tokens, API keys) before logging. On logout, sessions on other devices are revoked through the `logout-all` endpoint. The portal carries 228 Vitest tests across 76 suites, covering pure functions (`query-builder`, `curator-filters`, `curator-helpers`, `redact`, `safe-log`, `utils`), hooks (`useAiCurator`, `useAiCuratorStats`), configuration (`role-config`) and React components (`video-player`, `save-to-watchlist`, the legal pages, the studio pages). CI additionally runs `tsc --noEmit` to catch type errors before the build.

## Database Design

### Relational Database with Prisma

All business data resides in PostgreSQL, defined by 21 Prisma models in `backend/prisma/schema.prisma`. Prisma acts as the abstraction layer: every query is written against a typed API such as `findMany`, `upsert` or `createMany` and compiled into parameterised SQL, which eliminates the possibility of injecting raw SQL from application code. Fig. 10 illustrates the 5-domain relational architecture across all 21 Prisma models.

*Five-domain relational database architecture diagram (21 Prisma models on Supabase PostgreSQL).*

*The 21 Prisma models with their scalar and relation field counts.*

| Model | Business role | Scalar | Relation |
|---|---|---|---|
| `User` | User account + RBAC | 14 | 13 |
| `RefreshToken` | Rotating login sessions | 6 | 1 |
| `PasswordResetToken` | Password reset (1h TTL) | 7 | 1 |
| `ProductionTag` | Content tag per production | 5 | 0 |
| `LiveChannel` | Broadcast channel | 29 | 5 |
| `LiveEvent` | Live event / programme | 32 | 7 |
| `Recording` | Published recording | 42 | 5 |
| `Podcast` | Podcast collection | 12 | 3 |
| `PodcastEpisode` | Podcast episode | 12 | 1 |
| `Follow` | Channel follow | 4 | 2 |
| `Comment` | Comment | 11 | 6 |
| `CommentLike` | Comment like | 4 | 2 |
| `Reaction` | Realtime reaction | 6 | 3 |

| Model | Business role | Scalar | Relation |
|---|---|---|---|
| `WatchHistory` | Watch history | 8 | 2 |
| `Notification` | In-app notification | 13 | 2 |
| `AuditLog` | Audit log (gRPC) | 11 | 0 |
| `Watchlist` | Watchlist | 7 | 1 |
| `AiCuratorReport` | AI Curator report | 15 | 1 |
| `DeviceToken` | FCM device token | 8 | 1 |
| `NewsletterSubscriber` | Newsletter subscriber | 14 | 1 |
| `StreamDonation` | Livestream donation | 14 | 3 |

## Data-Domain Grouping

The models are organised into five business groups:

- **Authentication and identity.** `User` stores account details together with the role, `RefreshToken` stores rotating login sessions, and `PasswordResetToken` stores password-reset codes with a one-hour lifetime.
- **Broadcast content.** `LiveChannel` is the channel, `LiveEvent` is a scheduled programme and active live broadcast, `Recording` is a published recording and `ProductionTag` is a content tag attached to a production.
- **User interaction & Live Monetization.** `Comment`, `CommentLike`, `Reaction`, `Follow`, `WatchHistory`, `Watchlist` and `StreamDonation` (Super Chat live donations) describe the full spectrum of viewer behaviour.
- **Operations and auditing.** `AuditLog` receives records from the gRPC microservice, `Notification` stores in-app notifications, `DeviceToken` stores FCM tokens and `NewsletterSubscriber` stores newsletter subscribers.
- **AI assistance.** `AiCuratorReport` stores AI curation output together with its prompt and model version provenance.

## Relationships and Constraints

Relationships are expressed with standard Prisma foreign keys, and each carries an index matched to its hot-path query. Three constraints deserve attention:

1. **User deletion does not destroy the audit trail.** `RefreshToken.user` uses `OnDelete: SetNull`, so deleting an account invalidates the sessions while preserving the audit trail.
2. **A unique key on content tags.** `ProductionTag` carries a unique constraint on the programme-and-tag pair, so the same tag cannot be attached twice.

3. **Time-axis indexes.** EPG queries always filter by `scheduledAt` and `channelId`, so those columns are indexed; watch-history queries filter by `userId` and `watchedAt`.

## Normalisation and Design Decisions

The model is normalised to third normal form: attributes depend only on the primary key and no non-key attribute depends transitively on another. Two design decisions deserve discussion:

- **EPG structure normalisation.** Rather than storing the filled 24-hour grid, the system stores real events with content tags and recomputes the grid on demand. This keeps the source data truthful and lets schedules change without data migration, at the cost of computation at read time — a sound trade-off when most writes are editorial and most reads are schedule viewing.
- **AI reports are not stored as a JSON blob.** `AiCuratorReport` exposes separate columns for each important field (confidence score, broadcast-suitability class, compliance flags) so they can be filtered and aggregated in SQL, keeping only the free-form analysis in a dedicated text column.

## Schema Versioning

Every schema change is managed through `prisma migrate`, which generates SQL files under `backend/prisma/migrations` that are kept under version control. CI runs `prisma migrate deploy` against a temporary PostgreSQL 15 service before building the application, so an incompatible schema is detected before any production deployment.

# Testing and Evaluation

## Test Strategy

The project applies a layered test strategy, with unit testing of the business rules taking priority because that layer holds the most subtle logic. A distinctive property of this project is that the *entire back-end suite runs without a database*: each spec file injects a fake Prisma client, so the test cycle is short and fully deterministic.

## Execution Results

GitHub Actions run 40630318005 started at 11:05:17 and completed at 11:13:45 ICT on 28 October 2026, commit edafb234. Its artifacts record 468/468 passing tests in job scope (163 backend, 228 web, 77 mobile). The mobile workflow selected data/logic/presentation; 14 tests outside that scope were not included. The separate Playwright record reports 6/8.

*Automated test results across the three suites.*

| Test suite | Command | Total | Passed |
|---|---|---|---|

| Test suite | Command | Total | Passed |
|---|---|---|---|
| Backend (node:test) | `npm test` | 163 | 163 |
| Frontend (Vitest) | `npx vitest run` | 228 | 228 |
| Mobile (flutter test) | `CI selected directories` | 77 | 77 |
| **Total** | | **468** | **468** |

## Back-End Test Suite

The back-end suite uses Node.js 20's built-in `node:test` runner together with a custom runner (`backend/test/runner.js`) that compiles the specs with `tsc` and executes them sequentially. Choosing the built-in runner over Jest or Vitest removes every third-party dependency from the test path, shrinking the failure surface and keeping the CI pipeline stable.

*Back-end test distribution per spec file.*

| Tệp spec | Tests passed | Pass rate |
|---|---|---|
| `ai-curator.service.spec.ts` | 19 | 100% |
| `epg-filler.spec.ts` | 18 | 100% |
| `live-studio.service.spec.ts` | 18 | 100% |
| `auth.service.spec.ts` | 17 | 100% |
| `ai-curator.e2e.spec.ts` | 13 | 100% |
| `watchlist.service.spec.ts` | 11 | 100% |
| `ingest-sources.spec.ts` | 11 | 100% |
| `ai-curator-stats.service.spec.ts` | 10 | 100% |
| `channels.service.spec.ts` | 7 | 100% |
| `programs.conflict.spec.ts` | 7 | 100% |
| `programs.counter.spec.ts` | 8 | 100% |
| `audit-logger.service.spec.ts` | 8 | 100% |
| `epg-timeline.spec.ts` | 5 | 100% |
| `search.service.spec.ts` | 4 | 100% |
| `users.service.spec.ts` | 4 | 100% |
| `billing.service.spec.ts` | 3 | 100% |
| **Tổng** | **163** | **100%** |

Two spec files deserve particular mention because they directly protect the core algorithms. `programs.conflict.spec.ts` exercises the conflict detector across all three overlap forms, the `NULL`-duration case and the self-exclusion case used when updating an event. `epg-filler.spec.ts` verifies the completeness of the 24-hour grid, the determinism of the seeded selection, the repeat limit and the episode-splitting behaviour.

## Web Test Suite

228 Vitest tests in 76 suites cover three kinds of subject: pure functions, React hooks and interface components. Pure functions are tested at their boundaries (empty strings, case-insensitivity, invalid parameters); hooks are tested across loading, success and error states; components are tested with `React Testing Library` using user-facing queries rather than raw DOM structure assertions.

## Mobile Suite and End-to-End Tests

The Flutter test tree contains 91 tests. Workflow run 40630318005 executed 77 tests from `test/data`, `test/logic` and `test/presentation`; `test/core` and `test/widget_test.dart` are outside this CI command.

Alongside it, a Playwright-based end-to-end suite executes eight scenarios in Chromium: the login form rejects an invalid email, a wrong password surfaces an error toast, the EPG page renders without crashing, the channels page displays content, the search page accepts input and renders results, the root URL shows a public landing surface, and two scenarios verify unauthenticated redirect behaviour.

Note on the two remaining E2E failures

In the most recently documented run, six of eight scenarios passed. The STAFF login did not leave `/login` within the wait timeout; the unauthenticated `/studio` check received HTTP 401 instead of the redirect expected by its assertion. The report attributes the first to a missing seeded account. Raw traces, exact failure times, fix commits and a passing regression run are not retained. Both are documented in the accompanying `TEST_EXECUTION_REPORT.md`.

## Quality Assessment

Based on these results, the system reaches a reliability level appropriate for an educational deployment. The clearest strength is coverage of the business logic: the two riskiest algorithms (conflict detection and EPG filling) are tested at their boundaries rather than only on the happy path. The area needing improvement is end-to-end coverage, currently limited to eight browser scenarios and excluding the live-broadcast scenarios that would require dedicated hardware and MediaMTX infrastructure.

# Deployment and Continuous Integration

## Deployment Architecture

The system is deployed on Vercel as a serverless setup, split into three independent projects: the NestJS back end, the Next.js portal and a static service. All three production URLs have been verified to return HTTP 200.

*Production endpoints and verification status as of 28 October 2026.*

| Service | URL | Status |
|---|---|---|
| omnicast.vercel.app | https://omnicast.vercel.app | HTTP 200 |
| omnicast-fe.vercel.app | https://omnicast-fe.vercel.app | HTTP 200 |
| omnicast-api.vercel.app | https://omnicast-api.vercel.app | HTTP 200 |

## The CI/CD Pipeline

The file `.github/workflows/ci.yml` defines five jobs running in parallel plus a quality gate. Three test jobs run independently on `ubuntu-latest` with Node.js 20, Java 17 and stable Flutter. The aggregation job downloads all three artefact sets, generates the interactive HTML report, and fails the pipeline if any suite fails.

*The five-job CI/CD pipeline. Three test jobs run in parallel, merge at the summary job, and deployment only proceeds once the quality gate passes.*

The deploy job runs only on the `main` branch on a `push` event; it uses `vercel-action` to push the back end and the portal, passing `NEXT_PUBLIC_API_URL` from a repository secret. Because the deploy job sets `continue-on-error`, a deployment failure does not destroy the test artefacts already produced.

## Environment Configuration

The environment template lives at `backend/.env.example` and covers four groups: the Supabase database connection with a password placeholder, the `JWT_SECRET` and token lifetimes, the third-party API keys (OpenAI, TMDB, YouTube, Twitch, Cloudinary), and the local MediaMTX configuration. The real `.env` file is never committed.

## Deployment Security

Eight application-layer security measures are in place: `Helmet` sets the HTTP security headers; `ThrottlerModule` caps traffic at 100 requests per minute; CORS uses an explicit allow-list together with Vercel-domain and private-network checks; `ValidationPipe` strips undeclared fields; `HttpExceptionFilter` normalises errors; passwords are hashed with 12-round bcrypt; refresh tokens are rotated and revocations are persisted; and the asynchronous audit write is wrapped in a non-awaited promise so an infrastructure failure cannot break a business operation.

## Load and Resilience Testing

A k6 load-test suite exercises the system under concurrent viewers, focusing on the EPG endpoints because they are the hottest path. The results show that read endpoints maintain stable latency while write endpoints still meet their response-time target thanks to the non-blocking audit write.

# Conclusion and Future Work

## Conclusion

This paper has presented OmniCast, a multi-platform digital broadcast scheduling, real-time LiveStream and delivery system developed by the four-member engineering team (**Nguyễn Bình An**, **Trần Thanh Sơn**, **Nguyễn Vĩnh Phú**, **Đồng Hoàng Nguyên**) at **FPTU Quy Nhon AI Campus — FU Cahata** under the mentorship of **Mr. Huỳnh Vang**. Its four principal technical contributions are demonstrated through the source code and real test runs: (1) a bi-directional LiveStream and Live Studio architecture combining WebRTC WHIP/WHEP, MediaMTX HLS and Socket.IO `/live`; (2) a broadcast-conflict algorithm implemented as a single raw SQL query that computes the effective end time; (3) a 24/7 EPG filler whose seeded selection guarantees reproducibility; and (4) an AI curation pipeline that issues a single consolidated model call, validates the output against a Zod schema, offers a heuristic fallback and records full version provenance.

The complete system — 19 NestJS modules, 140+ REST endpoints, 21 Prisma models, a Next.js 15 portal with 33 routes, a 31-screen Flutter mobile app and a gRPC audit microservice — has been deployed on Vercel. CI run 40630318005 passed 468 tests in its job scope: 163 backend, 228 web and 77 mobile tests.

## Known Limitations

Three known limitations should be stated. First, browser E2E test coverage can be expanded further for hardware WHIP publishing scenarios requiring a dedicated MediaMTX instance in CI. Second, the external ingestion layer currently polls YouTube, Twitch and RSS on cron intervals. Third, the filler algorithm selects by category and by seed rather than by individualised viewer history.

## Future Work

Four directions are prioritised for subsequent iterations:

1. **Recommender-driven scheduling.** Instead of selecting by category, the filler algorithm could use watch history and interaction data to propose content per channel while preserving determinism so the result remains testable.
2. **Scaling MediaMTX end-to-end testing.** Provision a MediaMTX instance in CI to test the WHIP publish and HLS playback paths automatically, and add browser scenarios for the STAFF and ADMIN roles.
3. **Improving the AI assistant.** Add multi-model evaluation, second-model output verification, and ranking of AI reports by editor feedback so the prompt can be tuned iteratively.

4. **Separating the data-access tier.** Move EPG queries into a read-only analytical store with covering indexes to separate read load from write load on the same database.

## Acknowledgements and Reproducibility Statement

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# Appendix A: REST Endpoint Inventory

The following table is generated automatically by analysing the 19 controller files in `backend/src`. It covers 140 endpoints across 19 modules; the base-path column shows the prefix to use when calling them.

| Module | Base path | GET | POST | PUT | PATCH | DELETE |
|---|---|---|---|---|---|---|
| programs/programs | /api/v1/programs | 10 | 11 | 0 | 2 | 2 |
| programs/live-studio | /api/v1/programs/studio | 5 | 11 | 0 | 0 | 0 |
| channels/channels | /api/v1/channels | 7 | 4 | 0 | 1 | 1 |
| users/users | /api/v1/users | 5 | 1 | 0 | 4 | 0 |
| auth/auth | /api/v1/auth | 1 | 8 | 0 | 0 | 0 |
| programs/ingest/ingest | /api/v1/programs/ingest | 1 | 8 | 0 | 0 | 0 |
| programs/social | /api/v1 | 4 | 4 | 0 | 0 | 1 |
| notifications/notifications | /api/v1 | 2 | 3 | 0 | 0 | 2 |
| ai-curator/ai-curator | /api/v1/ai-curator | 2 | 4 | 0 | 0 | 0 |
| audit-logger/audit-logger | /api/v1/audit-logs | 6 | 0 | 0 | 0 | 0 |
| watchlist/watchlist | /api/v1/me/watchlist | 2 | 2 | 0 | 0 | 1 |
| billing/billing | /api/v1/billing | 2 | 2 | 0 | 0 | 0 |
| cloudinary/cloudinary-upload | /api/v1/uploads/cloudinary | 1 | 2 | 0 | 0 | 1 |
| search/search | /api/v1/search | 4 | 0 | 0 | 0 | 0 |
| health/health | /api/v1/health | 3 | 0 | 0 | 0 | 0 |
| newsletter/newsletter | /api/v1/newsletter | 1 | 2 | 0 | 0 | 0 |
| programs/stream-health | /api/v1/programs/health | 1 | 2 | 0 | 0 | 0 |
| admin/admin | /api/v1/admin | 1 | 1 | 0 | 0 | 0 |
| programs/studio-catalog | /api/v1/programs/studio/catalog | 2 | 0 | 0 | 0 | 0 |

# Appendix B: Source-Code Scale

A summary of the source-code scale of the three deliverables, counted as physical lines per file. These figures quantify the real volume of work performed and underpin the maintenance-effort estimate.

*Source-code scale per component.*

| Component | Lines of code | Unit |
|---|---|---|
| Backend NestJS (`src/`) | 27,826 | file .ts |
| Backend test (`test/`) | 4,512 | file .ts |
| gRPC proto | 83 | file .proto |
| Audit-logger microservice | 382 | file .ts |
| Prisma schema | 683 | file .prisma |
| Frontend Next.js (`app/`, `components/`, `lib/`) | 51,742 | file .ts/.tsx |
| Mobile Flutter (`lib/`) | 53,922 | file .dart |
| Mobile test (`test/`) | 2,529 | file .dart |

# Appendix C: Repository Layout

```
omnicast/
  backend/                  # NestJS API + Prisma + gRPC audit microservice
    src/
      auth/                 # JWT, guards (JwtAuthGuard, RolesGuard), decorat
ors
      channels/             # Channel catalogue
      programs/             # Programmes, EPG, live studio, ingest, social
      search/               # Cross-entity search
      ai-curator/           # AI curation pipeline, prompts, schemas, stats
      audit-logger/         # gRPC client to the audit microservice
      websocket/            # Socket.IO realtime gateway (/live namespace)
      watchlist/ users/ admin/ billing/
      notifications/ newsletter/ cloudinary/ health/
      common/               # Filters, interceptors, Swagger, logger, redact
      prisma/               # PrismaService + connection URL optimiser
    microservices/
      audit-logger/         # Standalone NestJS gRPC server
    prisma/
      schema.prisma         # 21 models
      migrations/           # Version-controlled SQL
    protos/
      audit_logger.proto    # gRPC service definition
    test/                   # 16 spec files, node:test runner
  frontend/                 # Next.js 15 App Router + Tailwind
    app/                    # 33 routes (24 route folders)
    components/ lib/        # UI components, hooks, API clients
    __tests__/              # 76 Vitest suites
  mobile/                   # Flutter 3.x (see the PRM393 paper)
    lib/{core,data,logic,presentation}/
    test/{data,logic,presentation}/
  playwright-tests/         # 8 browser end-to-end scenarios
```

```
.github/workflows/ci.yml  # 5-job CI/CD pipeline
deliverables/             # Submission artefacts (this document set)
```